

A Lightweight LLM-Driven Multi-Agent Framework for Virtual Society Simulation with Memory Feedback

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Valentown — seven residents who plan, meet, remember and adapt

Two ways to simulate people

R Rule-based agent models

Fixed rules



Action

same input → the same behaviour, every day

No context · no planning · no memory

Macro patterns yes — human-like behaviour no

L LLM-driven agents

Prompt



LLM



Action

fluent — but one step at a time

Reactive: answers the prompt in front of it

Interaction ad hoc or random · no day structure

Missing: explicit structure for behaviour, interaction and feedback

One question, three requirements

How can we design a virtual society whose agents are structured, evolving and controllable over time?

1 Temporal structure

A day — not a stream of reactions

OUR ANSWER

Schedule-driven agents

2 Behavioural evolution

Yesterday changes tomorrow

OUR ANSWER

Memory feedback loop

3 Controllability

An explicit rule, plus a human override

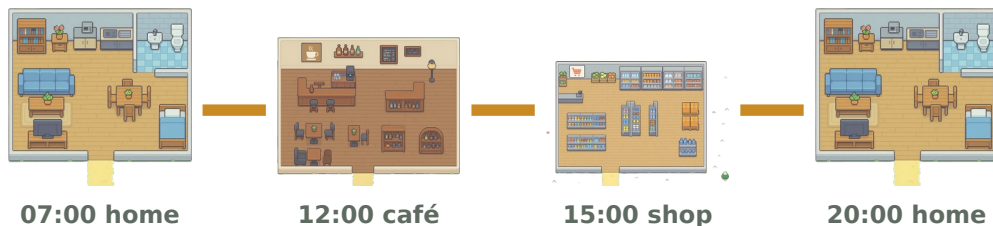
OUR ANSWER

Co-location trigger + human-in-the-loop

Plan the day, then let the town decide who meets

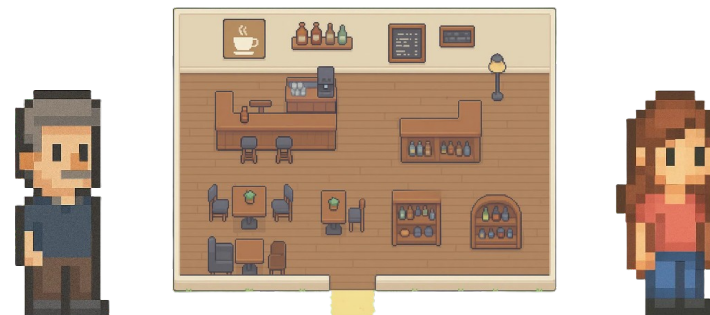
1 Schedule-driven day

identity + goals + memory → one day's plan



one generated day, illustrative

2 Co-location interaction



same place + same period → they talk

Deterministic when · generative what

the framework controls the dynamics — the model supplies the diversity

Memory is where the days connect

3 Memory-based feedback loop

WRITTEN IN

Plans

Conversations

Reflections

READ OUT

Planning

Dialogue

Reflection

Every entry: category · timestamp · importance score
Retrieval: recent + important · the store stays bounded

OPTIONAL: HUMAN IN THE LOOP

Feedback written into the same memory channel

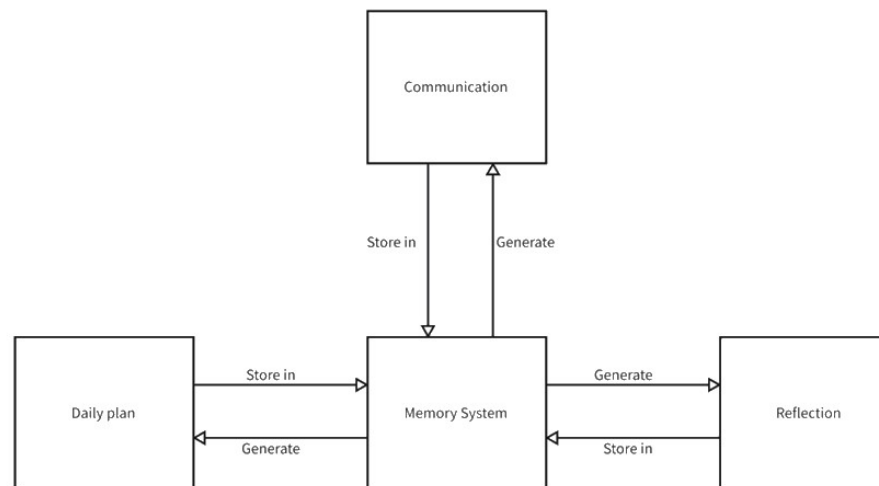


Fig. 3 Memory-centred feedback loop

The LLM stays stateless — all continuity lives in memory

One simulation day, end to end

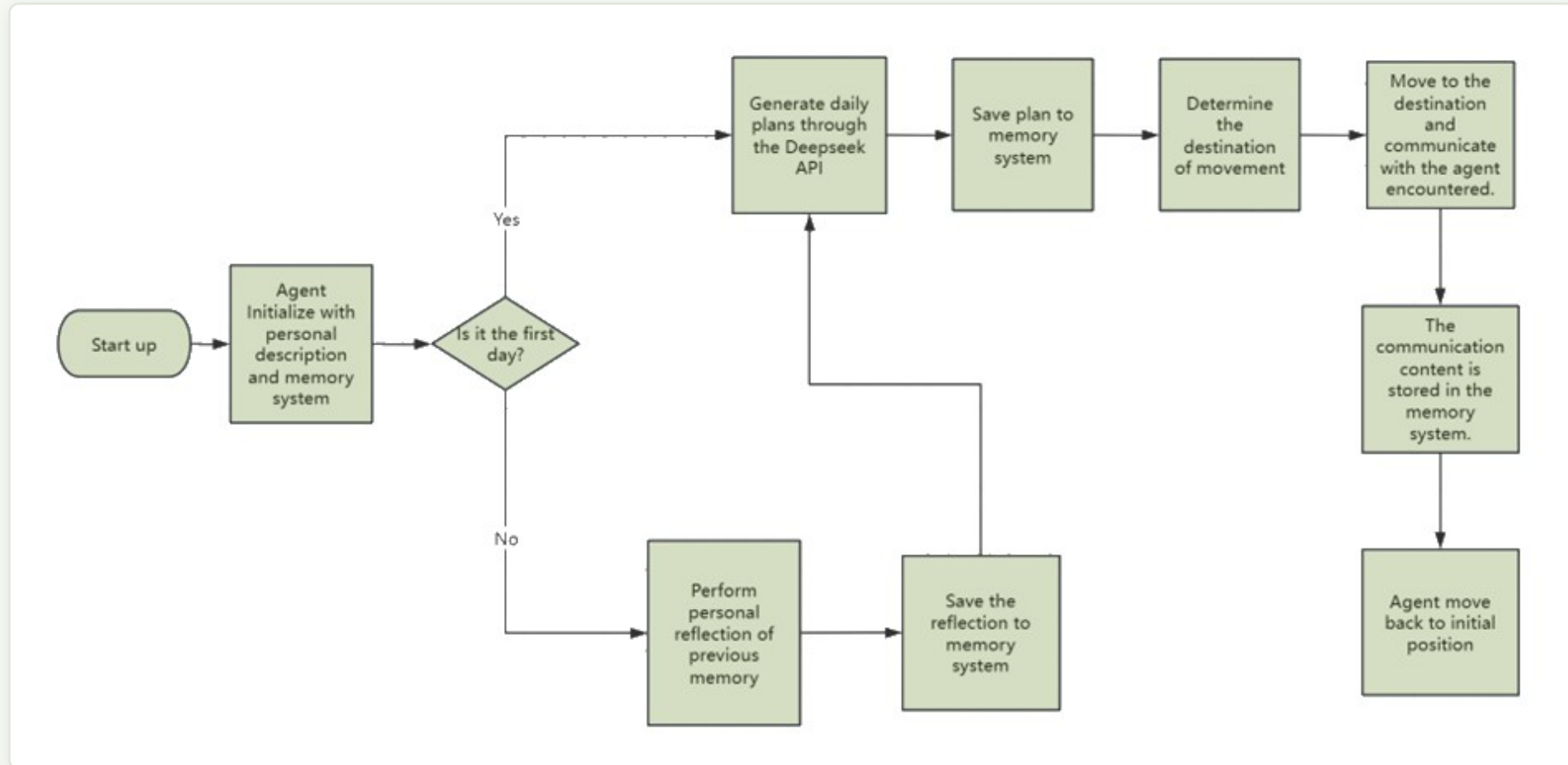


Fig. 2 · Day 1 plans from the profile — every later day reflects first, then plans

Valentown

RUNNING

7:28 AM

MORNING

Start

Pause

1x

2x

4x

RESIDENTS' NEEDS

Ron Parker

SUPERMARKET OWNER

FED 11

ENERGY 100

SOCIAL 45

Ella Parker

PHARMACY OWNER

FED 28

ENERGY 100

SOCIAL 22

Adam Harris

CHILD

FED 31

ENERGY 100

SOCIAL 43

Mia Thompson

FAMILY TEACHER

FED 21

ENERGY 100

SOCIAL 46

Emma Harris

MOTHER

FED 6

ENERGY 100

SOCIAL 42

Arthur Morgan

ARCHITECT

FED 96

ENERGY 100

SOCIAL 76

Gavin Harris

FATHER

FED 0

ENERGY 100

SOCIAL 29

SUPERMARKET OWNER

Ron Parker

STATUS

Preparing to move

DOING

Head to the kitchen and make a hearty breakfast to start the day.

AT

Ron home / Bed

Ron's Path

Control

SELF-REFLECTION

CONVERSATION LOG

No conversations recorded for this day yet.

a day planned overnight · two residents meeting at the same place · reflection reshaping tomorrow

EXPERIMENTS

What we ran, and what we looked for



Ron Parker

60 · supermarket owner



Ella Parker

58 · pharmacy owner



Emma Harris

30 · mother



Gavin Harris

32 · father



Adam Harris

7 · child



Mia Thompson

28 · family teacher



Arthur Morgan

29 · architect

Environment

predefined locations;
co-location enables interaction

Duration

multiple consecutive days,
a full cycle each day

Execution

deterministic triggers,
stochastic content

WE LOOK AT

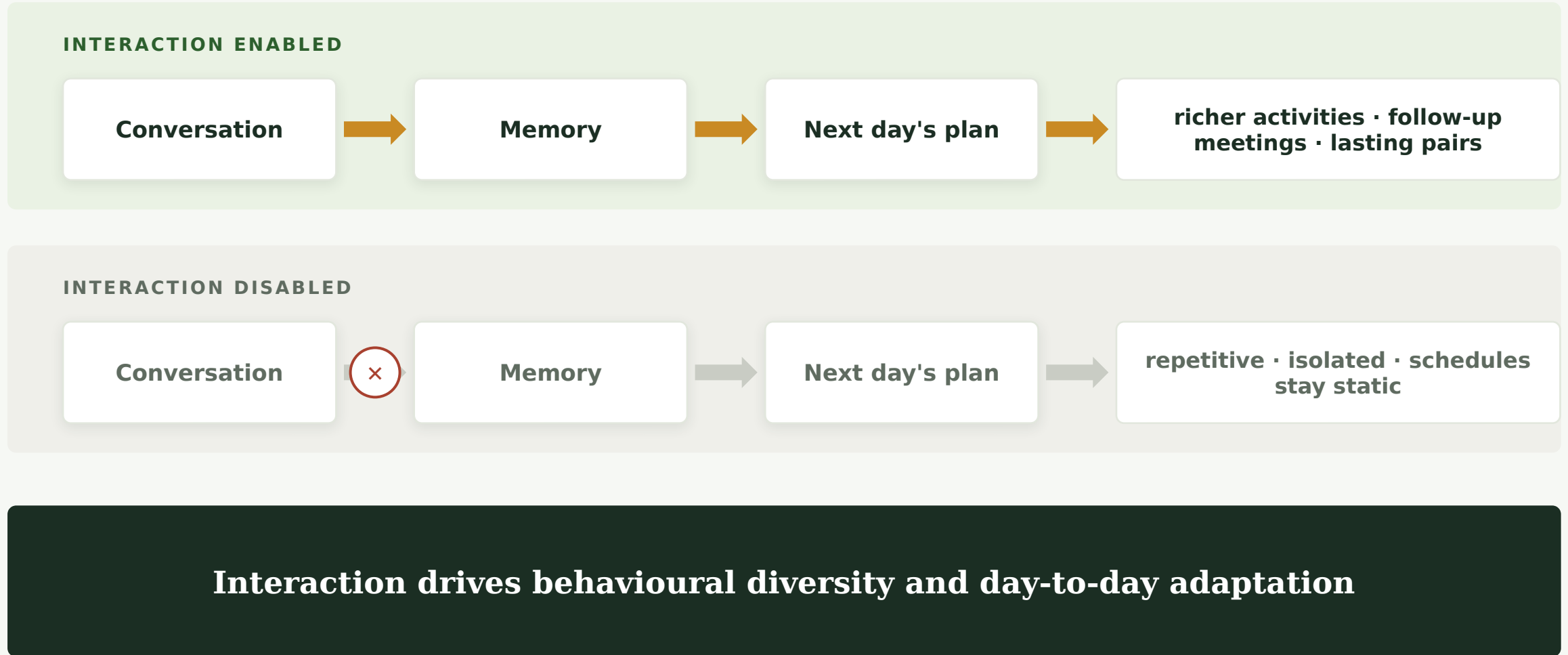
interaction → behaviour

memory → adaptation

emergent social structure

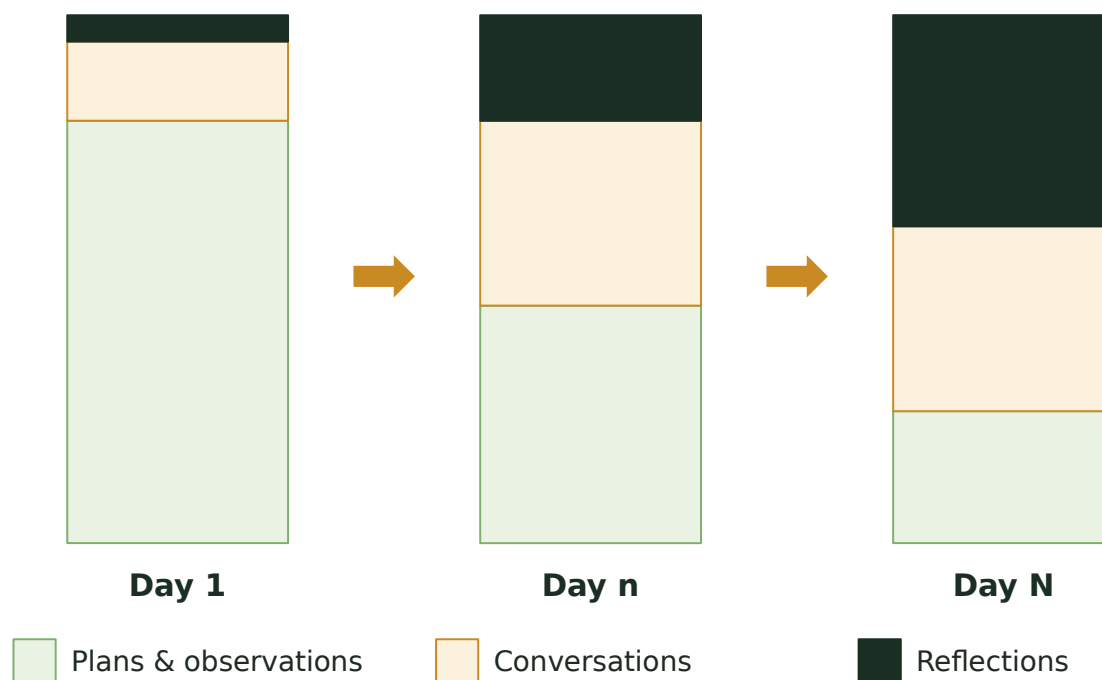
RESULT 1

Turn interaction off, and the town goes quiet



Memory changes shape as the days pass

What memory is made of



schematic — relative composition, not measured proportions

Bonds form on their own

Repeated meetings → persistent pairs → repeated co-location.

Nobody wrote a friendship rule.

Behaviour follows memory

Reflection-derived insight → goal-aligned task selection shifts across days.

Limit memory, and adaptation weakens.

Where this leaves us

What we showed

- 1 Planning, interaction and reflection linked by one memory loop
- 2 Importance-based memory + reflection → next-day behaviour
- 3 Consistent routines, lasting bonds, adaptation — no hand-written rules

AND WE OWN THE LIMITS

cost

scale

stochasticity

quantitative evaluation is the first thing we owe this work

Future work

- 1 **Quantitative evaluation**
network metrics, plan coherence, user studies

- 2 **Scale & real time**
larger populations, live reaction

- 3 **Deeper reasoning**
NPCs, education, embodied agents

Thank you

Questions are very welcome.

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Paper 108

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A backend that thinks, a frontend that shows

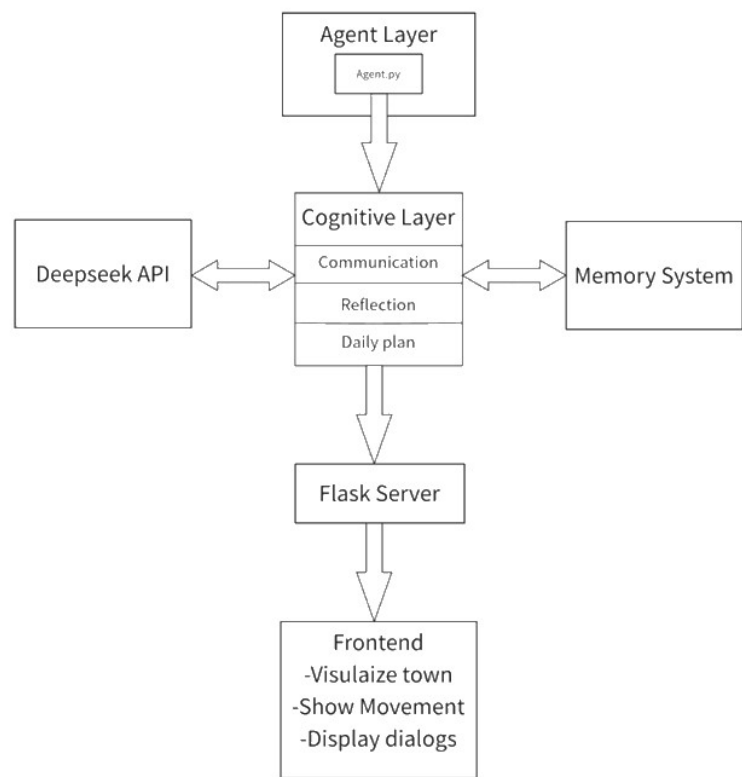


Fig. 1 Overview of the proposed framework

Backend · Python

Planning, dialogue, reflection, memory — the only part that calls the LLM

Frontend · Phaser

Town, movement and conversations, served over a REST API

Five modules

Agent Manager · Scheduler · Interaction Engine · LLM Interface · Memory Manager

Valentown: the framework, running



Valentown

homes · shop · pharmacy · café
· park

7 residents

own age, role, personality, goals

DeepSeek API

schedules · dialogue · reflection

Python + Phaser

pre-computed, replayed over
REST

What this buys us, and what it costs

Advantages

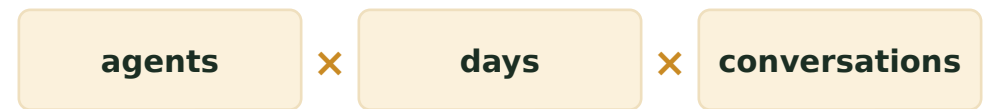
- 1 Lightweight & reproducible**
explicit pipeline, deterministic triggers
- 2 Controllable**
schedules, co-location rules, human feedback
- 3 Extendable**
every module swaps independently



any module can be swapped without touching the others

Limitations

- 1 Cost**
API calls scale with agents × days
- 2 Scale**
small population, simple town
- 3 Stochasticity**
runs vary — strict metrics are hard



every plan, conversation and reflection is an API call